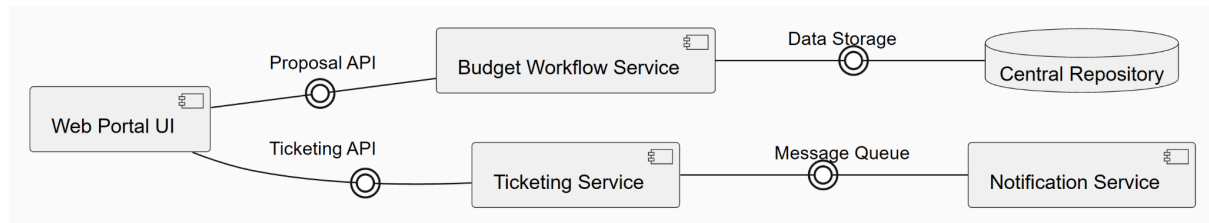


SE Lab 3

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Deliverable 1: Component Diagram



Deliverable 2: Written Justification

Architecture Selection: We chose Microservices

Architecture for the Student Club Event Ticketing & Budget Portal.

Reasons for Choice: The Microservices Architecture organizes the system into small, independently deployable services. The first reason for this selection is independent scaling; the portal handles two vastly different workloads. Processing a massive spike in student registrations for a popular quantum computing workshop requires heavy compute resources, whereas routing budget proposals is a low-traffic administrative task. The second reason is fault isolation. If the ticketing system crashes during a heavy load, the club heads and campus admins can still seamlessly access and approve budget proposals without system-wide downtime.

Security Advantage:

This architecture inherently supports strict, fine-grained access control boundaries. Because the Budget Workflow

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Service and the Ticketing Service are isolated components, internal financial data and faculty approval chains can be securely restricted to the campus intranet. External students scanning QR-code tickets at venue check-in points interact solely with the public-facing Ticketing Service, preventing any possibility of a compromised student account accessing university financial records.

Performance Benefit: Microservices allow specific components to be optimised for their unique latency requirements. By decoupling the ticket generation logic from the heavy database transactions of the budget module, the Ticketing Service can be allocated dedicated hardware resources. This ensures the QR-code ticket scanner API maintains the target latency of under 100 ms at venue check-in points, preventing bottlenecks even under simulated peak loads during large-scale technical events.